PhD in Neuroscience and Machine Learning Engineer.

Gamaliel Mendoza-Cuevas

Building, deploying, and monitoring production-grade deep learning models and scalable ML pipelines.

Executive Summary

Professional Overview

A Machine Learning Engineer with a PhD in Neuroscience, focused on taking models from prototype to production. I build end-to-end ML systems — training pipelines, model-serving endpoints, experiment tracking, and performance monitoring — across Databricks, Spark, and Python.
My work spans deep learning architectures (CNNs, Transformers, LLM fine-tuning) and classical machine learning, engineered for reliability, scalability, and measurable business impact in enterprise healthcare environments. I instrument models with MLflow for reproducibility and monitoring, and collaborate through Git, GitHub, and Bitbucket within cross-functional engineering teams.

Core Competencies & Professional Certifications 👉🔗🖱️

Machine Learning Engineering

Model Deployment · Model Serving · Real-Time Inference · ML Pipelines · MLOps · MLflow · Experiment Tracking · Model Monitoring · Model Lifecycle · Reproducibility · Scalable Systems

Deep Learning & ML

Deep Learning · Neural Networks · Computer Vision (CNNs) · NLP · Transformers · LLM Fine-Tuning · Tokenization (BPE) · Recommender Systems · Clustering (K-Means) · Predictive Modeling · Explainable AI

Languages & Tools

Python · PyTorch · TensorFlow · Scikit-learn · Transformers · MLflow · Spark · Databricks · SQL · R · MATLAB · Hadoop · Oracle · MySQL · Git · GitHub · Bitbucket

Data Engineering

ETL Pipelines · Feature Pipelines · Data Architecture · Data Warehousing · Data Quality · Batch Processing · Cloud & On-Premise Environments

Professional / Soft Skills

Remote Work · Cross-functional Collaboration · Executive Reporting · Data Visualization · Stakeholder Communication · Bilingual (EN/ES) · German (A2)

Domain

Healthcare Analytics · Healthcare Data · Neuroscience · Translational Research · Signal Processing

Professional Experience

Cotiviti

Aug 2025 – Aug 2026 | Remote, Mexico City, Mexico.

Associate Data Scientist — Applied Machine Learning & Model Deployment.

- Deployed and maintained model-serving endpoints for neural network classifiers, enabling real-time, scalable healthcare data intelligence.
- Delivered high-performance CNN-based classification solutions, enhancing predictive accuracy across mission-critical healthcare data assets.
- Led the fine-tuning of Transformer-based large language models, unlocking actionable insights from complex, unstructured healthcare data.
- Designed and implemented custom Byte-Pair Encoding (BPE) tokenization pipelines, optimizing downstream NLP performance on enterprise healthcare datasets.
- Instrumented experiment tracking, model performance monitoring, and reproducibility with MLflow, safeguarding model quality in production.
- Architected and operationalized end-to-end recommendation systems, driving measurable improvements in healthcare service delivery outcomes.
- Engineered K-means clustering-based recommendation engines to optimize resource allocation and elevate stakeholder value.
- Developed and deployed Python-based, ML-powered applications, translating advanced analytics into scalable, business-ready healthcare solutions.
- Built and scaled robust ETL pipelines across local and cloud (Databricks) environments, ensuring enterprise-grade data integrity and availability.
- Delivered a tiered Bronze/Silver/Gold (medallion) data architecture, establishing a single source of truth to accelerate organization-wide analytics.

BIOINVERT

Jan 2017 – Jun 2017 | Mexico City, Mexico

Data Analyst.

Automated quantitative analysis of medical imaging data and delivered statistical models published in the peer-reviewed literature.

- Conducted advanced CT imaging analysis to quantify bone density, supporting evidence-based clinical decision-making.
- Delivered statistical modeling of age-related bone density trends, generating actionable insights for longitudinal health research.
- Co-authored a peer-reviewed publication accepted in Veterinary Radiology and Ultrasound, reinforcing thought leadership in the field.

APREXBIO

Jan 2016 – Jan 2017 | Mexico City, Mexico

Research Intern.

Built signal-processing workflows for neurophysiological data and standardized benchmark values for downstream analysis.

- Executed analysis of visual evoked potentials.
- Achieved certification in MRI equipment operation and acquisition/analysis of anatomical imaging data.
- Co-authored a book chapter establishing standardized benchmark values for visual evoked potentials.
- Co-authored a peer-reviewed publication accepted in Experimental Gerontology.

Academic Credentials 👉🔗🖱️

PhD

Institute of Neurobiology | National Autonomous University of Mexico | Aug 2020 – Apr 2026 | Queretaro, Mexico

Neuroscience.

Doctoral research portfolio spanning computational modeling, signal processing, behavioral analytics, and neurophysiology, delivered within a top-tier Biomedical Sciences program.

- Delivered a peer-reviewed publication accepted in the Journal of Neurophysiology.
- Doctoral Thesis: Engineered a low-dose L-DOPA intervention strategy that measurably enhanced visual discrimination performance in a Parkinson's Disease model.
- Doctoral Advisor: Dr. Luis Alberto Carrillo Reid.
- Awarded a competitive doctoral scholarship by the Mexican Secretariat of Science, Humanities, Technology, and Innovation in recognition of research excellence.

Master

Institute of Neurobiology | National Autonomous University of Mexico | Aug 2018 – May 2020 | Queretaro, Mexico

Neurobiology.

Built and validated quantitative models of visual discrimination performance within a translational Parkinson's Disease model.

- Graduated with Honorific Mention and a 9.16/10.00 GPA, placing in the top tier of the program cohort.
- Awarded a competitive Master's scholarship by the Mexican Secretariat of Science, Humanities, Technology, and Innovation.

Language Proficiencies 👉🔗🖱️

English

C1 — Professional Working Proficiency (TOEFL iBT: 110/120).

German

A2 — Elementary Proficiency, actively progressing (A1 certified via Goethe-Zertifikat).

Spanish

Native / Bilingual Proficiency.

Selected Publications

Impaired visually guided behavior in a mouse model of Parkinson's disease

Mendoza-Cuevas, G., Perez-Becerra, J., Velazquez-Contreras, R., Calderon, V., Carrillo-Reid, L.

Journal of Neurophysiology, 2025 | DOI: [10.1152/jn.00307.2025](https://doi.org/10.1152/jn.00307.2025)

Visual evoked potentials in the Rhesus monkey

Mendoza-Cuevas, G., Hernández-Godínez, B., Ibáñez-Contreras, A.

Book chapter, 2017 | ISBN: 9781536110753

Computed tomography is a feasible method for quantifying bone density in Macaca mulatta

Solís-Chávez, S., Castillo-Rivera, M., Arteaga-Silva, M., Ibáñez-Contreras, A., Hernández-Godínez, B., Morón-Mendoza, A., Mendoza-Cuevas, G., Morales-Guadarrama, A., Sacristan-Rock, E.

Veterinary Radiology and Ultrasound, 2018 | DOI: [10.1111/vru.12624](https://doi.org/10.1111/vru.12624)

Electrical activity of sensory pathways in female and male geriatric Rhesus monkeys (Macaca mulatta), and its relation to oxidative stress

Ibáñez-Contreras, A., Hernández-Arciga, U., Poblano, A., Arteaga-Silva, M., Hernández-Godínez, B., Mendoza-Cuevas, G., Toledo-Pérez, R., Alarcón-Aguilar, A., González-Puertos, V., Konigsberg, M.

Experimental Gerontology, 2018 | DOI: [10.1016/j.exger.2017.11.003](https://doi.org/10.1016/j.exger.2017.11.003)

The effect of oxidative stress on brain electrical activity and its repercussions on sensory organization in geriatric Rhesus monkeys in captivity

Ibáñez-Contreras, A., Hernández-Godínez, B., Mendoza-Cuevas, G., Hernández-Arciga, U., Königsberg, M.

Book chapter, 2017 | ISBN: 9781536110753

Awards & Recognition

Second Place — 9th State Health Research Forum

Recognized for excellence in visual discrimination research within a Parkinson's Disease model • Querétaro, Mexico • 2024

Third Place — Latin American Brain Initiative (LATBrain) Design Competition

2021